IPL AUCTION SYSTEM

-DATABASE MANAGEMENT PROJECT

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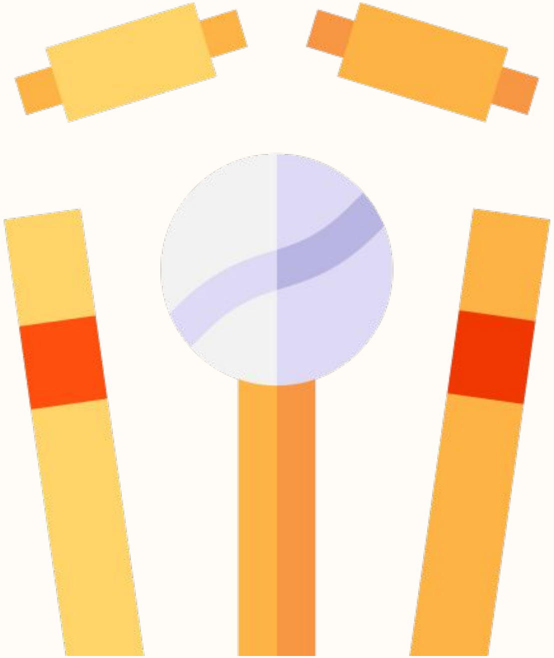


PROBLEM STATEMENT

The IPL (Indian Premier League) is one of the most popular cricket leagues in the world, featuring some of the best cricket players from across the globe.The IPL auction is an integral part of the league where the franchises bid for players to form their team.

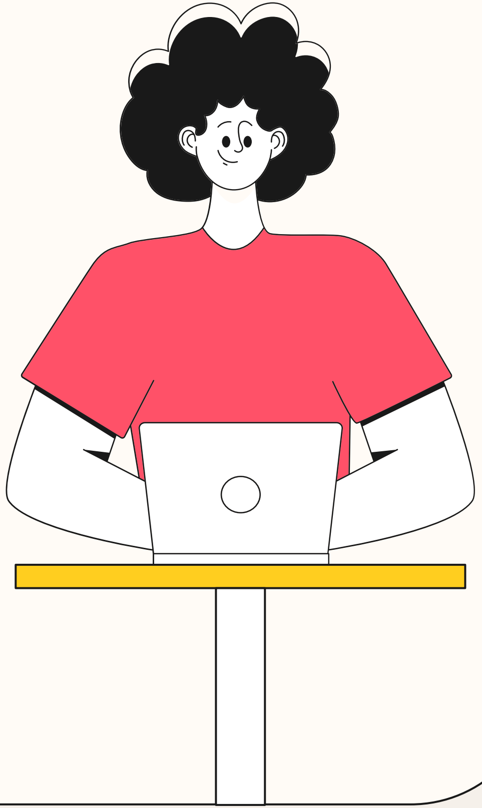
Design and develop a database for an IPL auction system using all the basic RDBMS concepts and execute all the CRUD operations. The database should be able to store and retrieve data related to the IPL auction process, including player information, franchise information, and auction results.

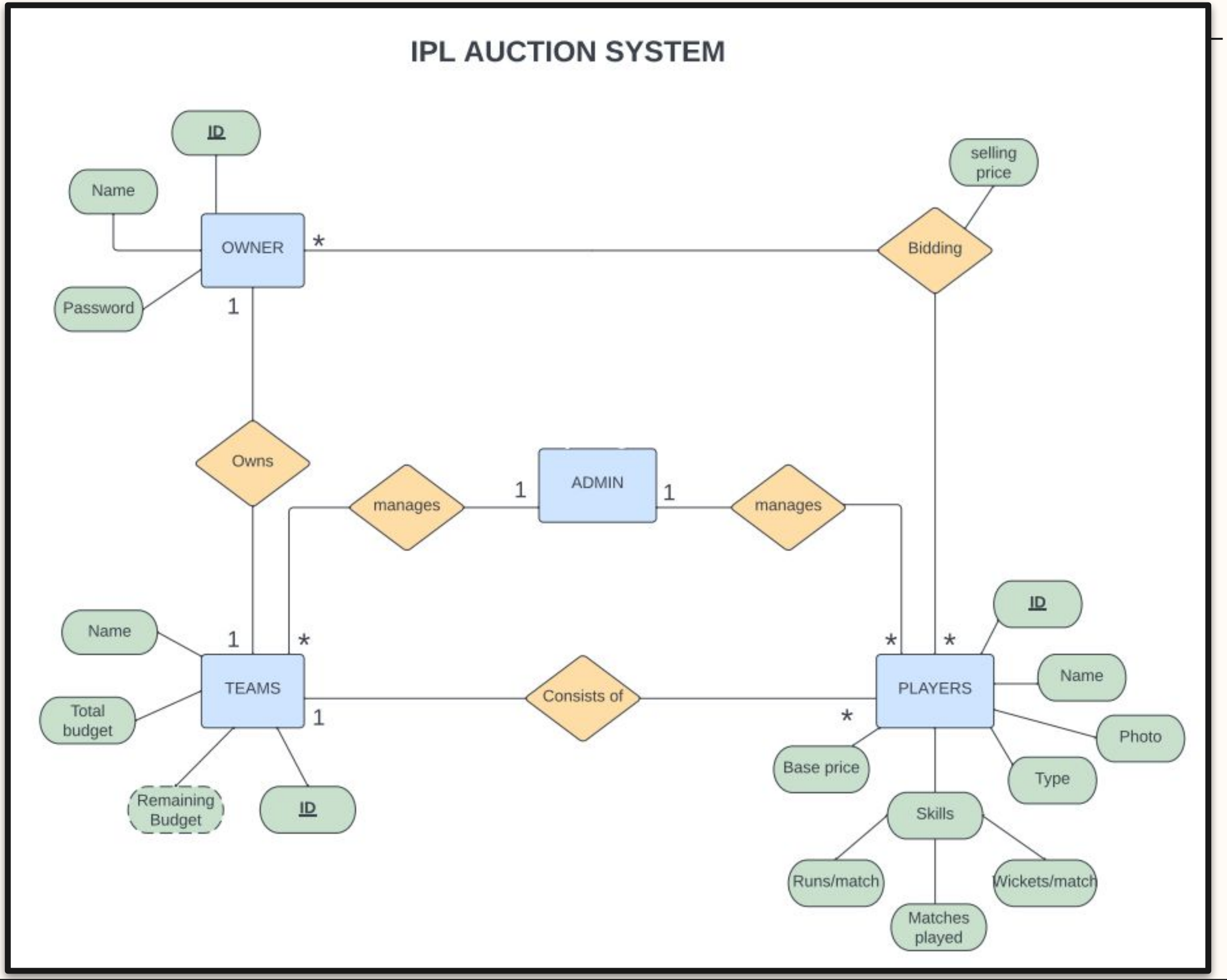
KEY FEATURES

1. **Team creation:** Users can create their own cricket teams by entering team name, owner name, and owner email address.
2. **Player management:** The system allows users to add, remove, and edit player details, such as name, age, playing position, and statistics.
3. **Auction management:** The system provides a platform for conducting player auctions, where teams can buy the players they need.
4. **Player statistics**: The system provides comprehensive statistics for each player, including their performance history, average runs, and wickets.
5. **User-friendly interface:** The system has an intuitive and user-friendly interface, making it easy for users to navigate and manage their teams.

DATABASE REQUIREMENTS

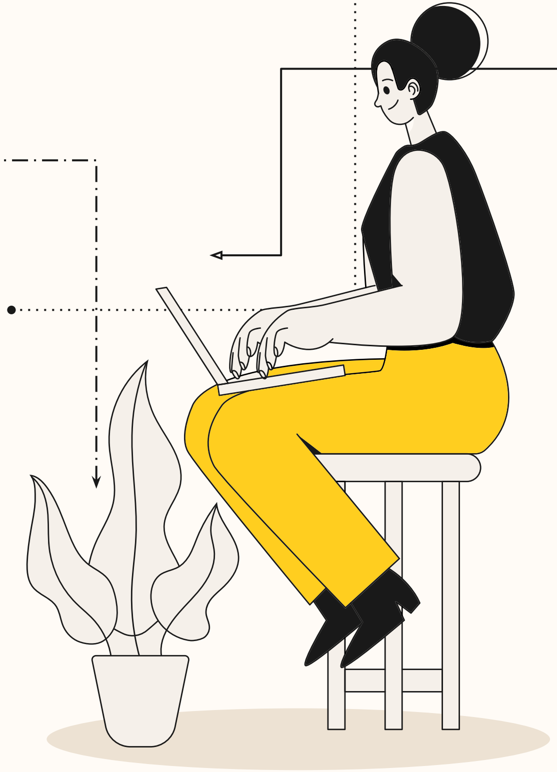
1. **Player table:** This table should store information about all the players who are up for auction, including their name, playing position, base price, nationality, and other relevant details like runs and matches.
2. **Franchise table**: This table should store information about all the franchises participating in the auction, including their name, owner, budget and other relevant details.
3. **Auction Result table**: This table should store information about the auction results, including the list of players who were sold, the final bid amount for each player, and the franchise that bought the player.




IPL AUCTION SYSTEM
ID
Name
OWNER
Password
*
1
selling price
Bidding
Owns
ADMIN
1
1
manages
manages
Name
1
*
TEAMS
1
Total budget
Remaining Budget
ID
Consists of
*
*
*
ID
PLAYERS
Name
Photo
Type
Base price
Skills
Runs/match
Wickets/match
Matches played

NORMALIZATION

- Normalization is the process of organizing the data in the database.
- Normalization is used to minimize the redundancy from a relation or set of relations. It is also used to eliminate undesirable characteristics like Insertion, Update, and Deletion Anomalies.



FIRST NORMAL FORM

First Normal Form (1NF) is the first step in the normalization process in SQL. Specifically, a table is in first normal form if:

1. Each table cell contains a single, indivisible value (i.e., atomic value).
2. Each column in a table has a unique name.
3. The order of rows and columns is irrelevant.
4. Each row in a table is unique; no two rows contain exactly the same data.

To convert a table that is not in 1NF into 1NF, you may need to split columns that contain multiple values into separate columns, or create additional tables to represent relationships between entities.

SECOND NORMAL FORM

Second Normal Form (2NF) is the second step in the normalization process in SQL. Specifically, a table is in second normal form if:

1. It is already in first normal form (1NF).
2. All non-key columns in the table are dependent on the entire primary key.

To bring a table that is not in 2NF into 2NF, we may need to create additional tables to represent relationships between entities, or split columns that contain multiple values into separate tables.

THIRD NORMAL FORM

Third Normal Form (3NF) is the next step in the normalization process in SQL. Specifically, a table is in third normal form if:

1. It is already in second normal form (2NF).
2. All non-key columns in the table are dependent only on the primary key.

In other words, a table should not contain any transitive dependencies, where a non-key column is dependent on another non-key column that is not part of the primary key.

To bring a table that is not in 3NF into 3NF, we may need to further split the table into smaller tables, or remove columns that contain redundant information.

NORMALIZED TABLES

Owner Table

ownerID	ownerName	ownerEmail	ownerPass
1	United Spirits	un@gmail.com	rcb
2	Ambani	MA@gmail.com	mi

NORMALIZED TABLES

Teams Table

teamID	teamName	totBudget
100	RCB	40
101	MI	40

NORMALIZED TABLES

Ownership Table

ownerID	teamID
1	100
2	101

NORMALIZED TABLES

Players Table

playerID	playerName	playerType	playerNation	playerImage	basePrice
111	Virat Kohli	Batter	Indian	PNG	15
222	Shane Watson	Bowler	Foreign	PNG	11

NORMALIZED TABLES

Skills Table

playerID	avgRuns	avgWickets	matches
111	56	2	234
222	23	5	109

NORMALIZED TABLES

Sold_players Table

playerID	teamID	sellingP
111	100	17
222	101	13

THANKYOU